

UBT706: ENZYME TECHNOLOGY

L	T	P	Cr
3	1	0	3.5

Course Objective: The objective of the course is to inform the students about isolation of novel enzymes, basic principles for optimization, modeling of the industrial use of the enzymes etc. in which both, free and immobilized enzymes play a role. Students will be able to implement both biochemical and engineering knowledge in order to design new and improve current enzymatic processes.

Syllabus

Introduction to enzymes: Enzymes-historical resume, Nomenclature and Classification, Biological roles, Isozymes, Marker enzymes, Overview of enzyme 3D-structure, active site, catalytic site, Specific activity, Enzyme activity and kinetics of free enzymes, K_m , V_{max} , and turn over number of enzymes and their significance, Modifiers of enzyme activity, enzyme activators, enzyme inhibitors, Overview of enzyme inhibition kinetics: competitive, non-competitive, and un-competitive, Kinetics of enzyme degradation, Enzyme stability: pH, temperature.

Production of enzymes: Sources of industrial enzymes (natural & recombinant), Screening for new and improved enzymes, different methods of extraction, isolation and purification of commercially important enzymes, retailoring of enzyme, large-scale industrial enzyme production and downstream processing: Case studies (from literature), improvement of enzymes.

Techniques of enzyme immobilization: Immobilization- definition, Immobilization techniques: physical and chemical adsorption, matrix entrapment, encapsulation, cross-linking, covalent binding with examples, Advantages and disadvantages of different immobilization techniques, Overview of applications of immobilized enzyme systems, Enzymes in electrodes.

Kinetics of immobilized enzymes: Steady state analysis of mass transfer and biochemical reaction in enzyme reactors, Analysis of mass transfer effects of kinetics of immobilized enzyme reactions, Packed bed reactors, Film and pore diffusion effects on kinetics of immobilized enzyme reactions, Effectiveness factors of immobilized enzyme systems.

Application of enzymes: Industrial uses of enzymes, proteolytic enzymes in meat and leather industry, Clinical enzymology, Therapeutic enzymes, Diagnostic enzymes.

Laboratory Work (if applicable)

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Course Learning Objectives (CLO)

The students will be able to:

1. Explain mechanism and function of enzymes and assay enzyme reaction kinetics.
2. Produce, isolate and purify enzymes at lab/industry scale.
3. Design enzyme immobilization techniques based on the types of enzymes.
4. Comprehend and calculations involved in immobilized enzyme kinetics.
5. Explain the industrial applications of enzymes.

Text Books

1. Arora NK, Mishra J, Mishra V, Microbial Enzymes: Roles and Applications in Industries, Springer Singapore (2020).
2. Vogel A and May O (Ed.) Industrial Enzyme Applications, Wiley (2019).
3. Aehle, Enzyme in Industry: Production and Applications, Wiley-VCH (2007).

Reference Books

1. Bisswanger H, Enzyme Kinetics: Principles and Methods, Wiley-VCH (2017).
2. Dixon M and Webb MC, Enzymes, Longmans (1980).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include Assignments /Tutorials /Quizzes)	30

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